Periconceptional Folic Acid and Neural-Tube Defects

Randomized and systematic-review evidence shows periconceptional folic acid substantially reduces neural-tube-defect risk; this does not generalize to all anomalies or to intelligence.

DE: HIGH-NTD; LOW-OTHER-ANOMALIES-COGNITION | STATUS: STABLE | DOMAIN: PREGNANCY, NUTRITION | REPLICATION: VERI

WHAT WE KNOW (Narrowest Supported Conclusion)

Daily periconceptional folic acid (alone or with other vitamins) reduces neural-tube-defect-affected pregnancy relative to no folate. The Cochrane synthesis of five trials reports RR 0.31 (95% CI 0.17–0.58; 6,708 births; high-quality evidence). This is a structural embryonic outcome, not a general intelligence or "optimal development" claim.

EVIDENCE AND CAUSAL ASSESSMENT

- Landmark RCT: MRC Vitamin Study (1991) established prevention of NTD recurrence with periconceptional folic acid.

- Independent first-occurrence RCT: Czeizel & Dudás (1992), NEJM, Hungarian periconceptional multivitamin trial; no NTD in the folic-acid arm versus six in the control arm among women without a prior NTD pregnancy.

- De-Regil et al. (2015) Cochrane review: five RCTs, 6,708 births, RR 0.31 (0.17–0.58) for NTDs; recurrence RR 0.34 (0.18–0.64). Dose subgroup (0.4 mg vs higher) did not change the conclusion. Cochrane new_version on this record is a successor review, not a retraction.

- WHO antenatal-care guidance recommends daily iron and folic acid as a public-health measure; that guideline is adjacent framework, not this card's identification.

EFFECT SIZE AND PERSISTENCE

Large relative-risk reduction for NTD. Persistence is binary at the structural level (defect present or not). There is no license to carry this effect size into IQ, school achievement, or all birth defects.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a for NTD. Baseline fortification and population folate status change the absolute benefit. Trials are older and often recurrence-focused; first-occurrence evidence is now represented by Czeizel & Dudás as well as the Cochrane pool. Adherence, timing (must cover neural-tube closure), and access are implementation issues, not reasons to doubt the core RCT result.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean perfect supplementation guarantees typical development, that an NTD implies parental nutritional failure in every case, or that folic acid is a general brain-optimization tool.

WHAT ACTION IS REASONABLY JUSTIFIED

Population fortification and periconceptional folic acid as recommended antenatal care are justified on NTD-prevention grounds. Supplement maximalism and cognitive-enhancement framing are not.

VERIFIED SOURCES & DOI IDENTIFIERS

- De-Regil et al. (2015), Cochrane Database of Systematic Reviews
- MRC Vitamin Study Research Group (1991), Lancet
- Czeizel & Dudás (1992), New England Journal of Medicine